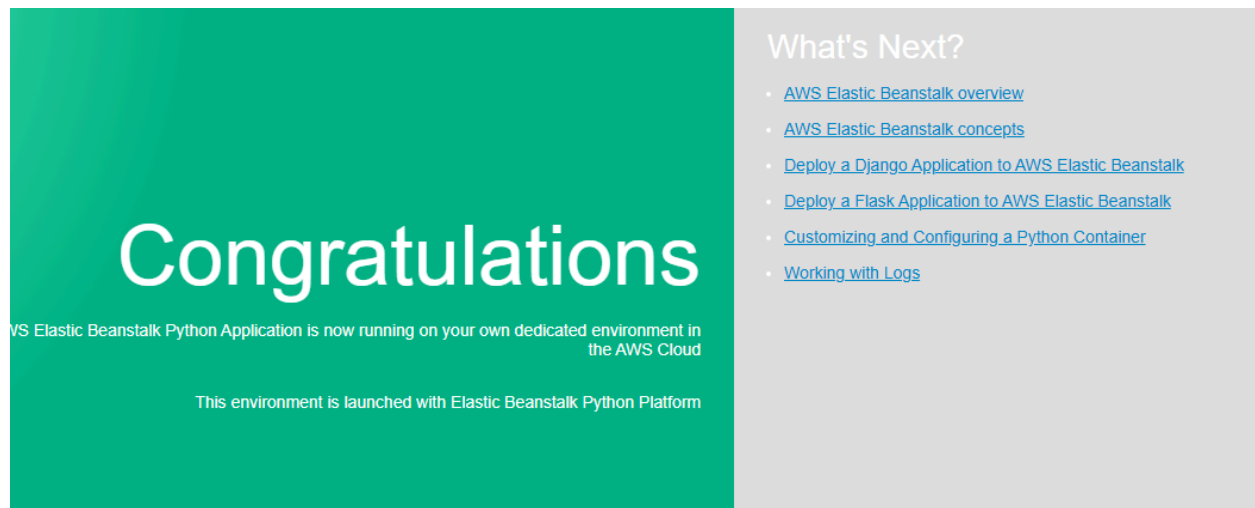


- Take a screenshot showing it has been brought up successfully



- Take a screenshot of the replacement VM being started.

Instances (3) [Info](#) Last updated less than a minute ago  [Conn](#)

All states ▾

☐	Name 	Instance ID	Instance state 	Instance type 	Status check	Alarm status	Availability Zone
☐	Eb-hello-env	i-05984d25a0279b63f	 Running  	t2.small	 2/2 checks passed	View alarms +	us-east-1c
☐	Eb-hello-env	i-0016d243aa8d4051f	 Terminated  	t2.small	–	View alarms +	us-east-1f
☐	Eb-hello-env	i-0de8d694d3a3ce0fc	 Running  	t2.small	 Initializing	View alarms +	us-east-1f

- Take a screenshot of the Guestbook including the URL with the entry in it.

Guestbook

Sign [here](#)

Entries

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 04:11:05.517491
Hello DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 04:39:00.827050
Hello Docker DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:30:27.282449
Hello Elastic Beanstalk!

- Take a screenshot of them.

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:sort=instanceState

Search

[Alt+S]

United States (N. Virginia)

voclabs/us

<

Instances (6) Info

Last updated less than a minute ago

Connect

Instance state

Acti

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	guestbook-...	i-01389c2c683bfc49	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1a	ec2-18-204-212-181.co...
☐	guestbook-env	i-069c3dba864de447d	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1b	ec2-100-24-37-228.co...
☐	guestbook-env	i-099da4f80a95a47d9	Running	t3.micro	3/3 checks passec	View alarms +	us-east-1c	ec2-54-242-162-153.co...

- Take a screenshot of the output that includes the URL in the address bar for your lab notebook.

Guestbook

[Sign here](#)

Entries

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 07:19:17.908081+00:00
Hello Compute Engine!

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 06:53:06.106613+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:41:06.371140+00:00
Hello Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:48:46.343697+00:00
Hello App Engine!

Joseph Gafron <joeygafron@gmail.com>
signed on 2025-02-06 07:19:02.526088+00:00
Hello Cloud Shell!

- Take a screenshot of them.

Instances

☐	ID 	QPS 	Latency 	Requests	Errors	Memory	Start Time	Availability
☐	 00ca3695d257486f3ac0465aa8...	0.067	55 ms	5	0	107.5 MB	Feb 11, 2025, 5:48:11 PM	Resident
☐	 00ca3695d2d38ca863ce61aee...	0.067	92 ms	5	0	107.4 MB	Feb 11, 2025, 5:48:11 PM	Resident

- Take a screenshot of the proxy and its results including the URL containing your OdinID

← ↻ 🔒

https://8000-cs-774697995690-default.cs-us-west1-wolo.cloudshell.dev/jgafron?url=https://oregonctf.org

Toggle navigation

Portland State University Computer Science
Oregon CTF

Capture-the-Flag security games and codelabs

Ones we've developed:

- Computer Systems Programming (CS 205) [CTF](#)
- Malware Reverse Engineering (CS 492) [CTF](#)
- angr Symbolic Execution (CS 492) [CTF](#)
- Cloud Security (CS 430/495) [Thunder CTF](#)
- Fuzzing (CS 492) [codelab](#)
- Divergent Cryptography and Security (CyberPDX camp) [CTF](#)

Ones we like to teach from:

- bandit (Linux tools) [CTF](#)
- natas (Web Security) [CTF](#)
- PortSwigger (Web Security) [CTF](#)
- OWASP Damn Vulnerable NodeJS Application (Web Security) [CTF](#)
- flaws.cloud (Cloud Security) [v1](#) | [v2](#)
- CloudGoat (Cloud Security) [exercises](#)
- Microcorruption (Reverse Engineering) [CTF](#)
- CryptoPals (Cryptanalysis) [CTF](#)

Portland State's CTF Slack channel [here](#)

Resources

Some recommended resources include:

- Download a Windows XP VM with IDA Pro Free installed [here](#)
- Or download IDA Pro Free [here](#)
- Download a Linux OS Box [here](#)
- PSU's CS 205 Computer Systems Programming [course](#)
- PSU's CS 430 Internet, Web, and Cloud Systems [course](#)
- PSU's CS 495 Web and Cloud Security [course](#)
- PSU's CS 492 Malware Reverse Engineering [course](#)
- PSU's CS 410 Blockchain Development and Security [course](#)

- **What is the security advantage of passing in the secret proxy route as an environment variable?**

Well if it is an environment variable rather than just a regular route, it protects against users accessing parts of our application that they aren't meant to access. Tools like dirsearch and gobuster will not be able to see our routes.

- **Take a screenshot of the image in the registry that shows the size of the container for your lab notebook.**

OVERVIEW	PULL	MANIFEST	FILES	ATTACHMENTS
Format	Docker			
Media type	application/vnd.docker.distribution.manifest.v2+json			
Project	cloud-gafron-jgafron			
Location	us-west1 (Oregon)			
Repository	secret-proxy-repo			
Image	secret-proxy			
Digest	sha256:b27f41b0e30aaebbbd2c855b61aea74dbac5dc2e047dd4fe6cb7fd7abcbc79e3			
Virtual size	53.5 MB			
Built	Feb 11, 2025, 9:30:50 PM			
Created	Feb 11, 2025, 9:31:00 PM			
Updated	Feb 11, 2025, 9:31:00 PM			
Tags	latest			
Subject digest	—			
Artifact type	—			

- Take a screenshot of it that includes the proxy URL for your lab notebook.

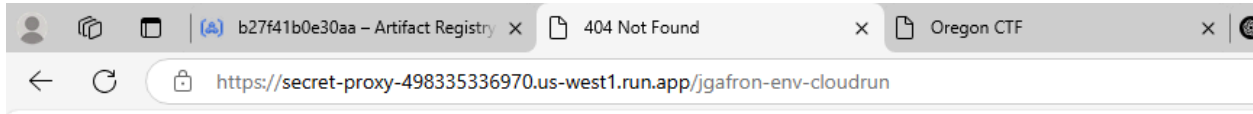
b27f41b0e30aa – Artifact Registry
secret-proxy-498335336970.us-west1
Oregon CTF

<https://secret-proxy-498335336970.us-west1.run.app/jgafron-env-cloudrun>

Proxy

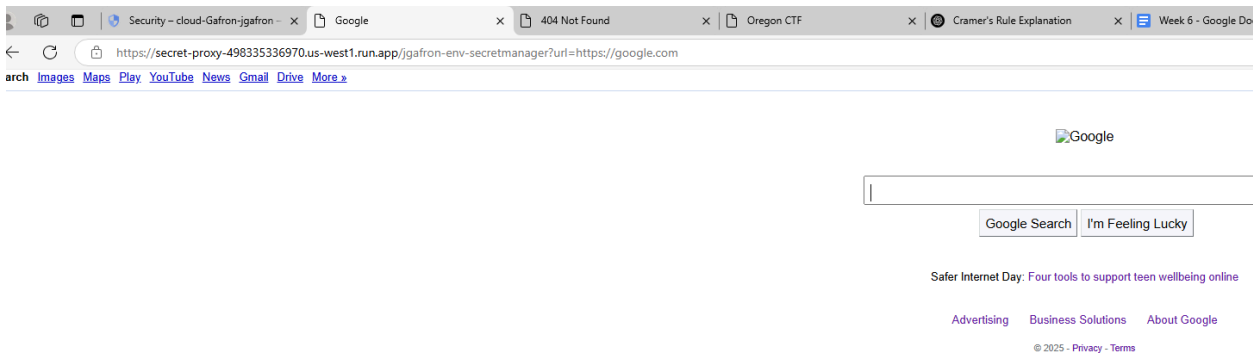
Enter URL to access by proxy:

- Take a screenshot of the error page that includes the proxy URL for your lab notebook.



Not Found

The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.



- **Identify the vulnerability in your lab notebook that Google has prevented.**

Basically Google prevented a request forgery from the user. Our proxy route is capable of retrieving a web page for the user. By entering those specific URLs, instead of retrieving a webpage for the user, our application would've retrieved metadata that only we should have access too. This metadata could've provided credentials for IAM roles that would have access to sensitive information, and could have granted the user access to things like data buckets.

Show that your image was uploaded to your account on [Docker Hub](#).

← ↻ <https://hub.docker.com>

New [More Docker. Easy Access. New \\$](#)

 **dockerhub** [Explore](#) [Repositories](#) [Organizations](#) [Usage](#)

▼ [All content](#)

Name	Last Pushed ↑	Contains	
jgafron/aws_gb	less than a minute ago	IMAGE	F
jgafron/gcp-gb	6 days ago	IMAGE	F
jgafron/hw3small	6 days ago	IMAGE	F
jgafron/hw3large	6 days ago	IMAGE	F
jgafron/helloalpine	8 days ago	IMAGE	F
jgafron/helloubuntu	8 days ago	IMAGE	F

- Take a screenshot of the DNS name of the guestbook-lb load balancer for your lab notebook

▼ Details

Load balancer type Application	Status ✔ Active	VPC vpc-0573a16596ef24f82 	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDTRQ7X7K	Availability Zones subnet-004cecaeed0637fe3  us-east-1a (use1-az2) subnet-050fad51309650023  us-east-1b (use1-az4) subnet-079ce4ff2b3299336  us-east-1f (use1-az5) subnet-0bf567acc3a6887bd  us-east-1d (use1-az1) subnet-08a2938ce86d84507  us-east-1c (use1-az6) subnet-0009cc054c948b8b6  us-east-1e (use1-az3)	Date created February 11, 2025, 22:59 (UTC-08:00)
Load balancer ARN  arn:aws:elasticloadbalancing:us-east-1:213294383979:loadbalancer/app/guestbook-lb/e10a125593047156		DNS name Info  guestbook-lb-60618737.us-east-1.elb.amazonaws.com (A Record)	

- Take a screenshot of the Guestbook app running in a browser that includes the DNS name of the site.

Guestbook

Sign [here](#)

Entries

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 04:11:05.517491
Hello DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 04:39:00.827050
Hello Docker DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:30:27.282449
Hello Elastic Beanstalk!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 07:05:30.149379
Hello ECS!

Joseph Gafron <joeysgafron@gmail.com>
signed on 2025-02-06 05:11:37.872590
Hello Cloud9!

Joseph Gafron <joeysgafron@gmail.com>
signed on 2025-02-06 05:24:18.392830
Hello EC2!

- Take a screenshot that includes the output of the command and the time it took to execute.

✓ Successful: 09ad1661-d84e-47b8-8790-3519d00a2a76

Started on Feb 11, 2025, 5:46:49 PM

Steps	Duration	Build Summary		
✓ Build Su... 3 Steps	00:01:10			
✓ 0: fetch -type=M...	00:00:04	Build log	Execution details	Build artifacts
✓ 1: pre-bui... /bin/shi...	00:00:08	☐ Wrap lines		
✓ 2: build /cnb/life...	00:00:55	☐ Show newest entries first		
		↑ ↓ ⌕ Expand View raw ↗		
		<pre>1 starting build "09ad1661-d84e-47b8-8790-3519d00a2a76" 2 3 FETCHSOURCE 4 BUILD 5 Starting Step #0 - "fetch" 6 Step #0 - "fetch": Pulling image: us-west1-docker.pkg.dev/serverless-runti 7 Step #0 - "fetch": base_20241212_18_04_RC00: Pulling from serverless-runti 8 Step #0 - "fetch": 4c84e3366e18: Pulling fs layer 9 Step #0 - "fetch": 4c84e3366e18: Verifying Checksum 10 Step #0 - "fetch": 4c84e3366e18: Download complete 11 Step #0 - "fetch": 4c84e3366e18: Pull complete 12 Step #0 - "fetch": Digest: sha256:0f10e63c2af36a9f5557c008d4ac03c4056658bb 13 Step #0 - "fetch": Status: Downloaded newer image for us-west1-docker.pkg. 14 Step #0 - "fetch": us-west1-docker.pkg.dev/serverless-runtimes/utilities/g</pre>		

- Take a screenshot showing the container image and its virtual size

Google Cloud

cloud-Gafron-jgafron

artifact

050765f5d9f52

DELETE

SETUP INSTRUCTIONS

DEPLOY

REFRESH

to Digests for gcp_gb

us-west1-docker.pkg.dev > cloud-gafron-jgafron > guestbook-repo > gcp_gb > sha256:050765f5d9f52fb3713daec5351fc5fd527a00f7f2e3aa328636b3c4f1062fb1

OVERVIEW

PULL

MANIFEST

FILES

ATTACHMENTS

Format

Media type

Project

Location

Repository

Image

Digest

Virtual size

Built

Created

Updated

Tags

Subject digest

Artifact type

Docker

application/vnd.docker.distribution.manifest.v2+json

cloud-gafron-jgafron

us-west1 (Oregon)

guestbook-repo

gcp_gb

sha256:050765f5d9f52fb3713daec5351fc5fd527a00f7f2e3aa328636b3c4f1062fb1

1.2 GB

Feb 11, 2025, 11:22:05 PM

Feb 11, 2025, 11:24:48 PM

Feb 11, 2025, 11:24:48 PM

latest

—

—

Take a screenshot that includes the URL Cloud Run has created for your site

Guestbook

Sign [here](#)

Entries

Joseph Gafron <joeysgafron@gmail.com>
signed on 2025-02-06 07:19:17.908081+00:00
Hello Compute Engine!

Joseph Gafron <joeysgafron@gmail.com>
signed on 2025-02-06 06:53:06.106613+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:41:06.371140+00:00
Hello Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:48:46.343697+00:00
Hello App Engine!

Joseph Gafron <joeysgafron@gmail.com>
signed on 2025-02-06 07:19:02.526088+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:42:58.746814+00:00
Hello Docker Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 07:36:13.859830+00:00
Hello Cloud Run!

- What port do container instances listen on?

8080

- **What are the maximum number of instances Cloud Run will autoscale up to for your service?**

100

- **After downloading the file from the bucket, where is it stored?**

It is stored in a file, that is called temp_file_name

- **What class in the ImageMagick package is used to do the blurring of the file?**

The "Image" class does the blurring

- **What lines of code perform the blurring of the image and its storage back into the filesystem?**

Blur the image using ImageMagick.

```
with Image(filename=temp_local_filename) as image:
```

```
    image.resize(*image.size, blur=16, filter="hamming")
```

```
    image.save(filename=temp_local_filename)
```

```
print(f"Image {file_name} was blurred.")
```

```
# Upload result to a second bucket, to avoid re-triggering the function.
```

```
# You could instead re-upload it to the same bucket + tell your function
```

```
# to ignore files marked as blurred (e.g. those with a "blurred" prefix)
```

```
blur_bucket_name = os.getenv("BLURRED_BUCKET_NAME")
```

```
blur_bucket = storage_client.bucket(blur_bucket_name)
```

```
new_blob = blur_bucket.blob(file_name)
```

```
new_blob.upload_from_filename(temp_local_filename)

print(f"Blurred image uploaded to: gs://{blur_bucket_name}/{file_name}")
```

- Take a screenshot of the blurred image in the output bucket for your lab notebook

←

Bucket details

GO TO PATH

📁

jgafronsoredimages

Location

Storage class

Public access

Protection

us-west1 (Oregon)

Standard

Subject to object ACLs

Soft Delete

<

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

Folder browser

🔍

📁

jgafronsoredimages

⋮

Buckets > jgafronsoredimages

CREATE FOLDER

UPLOAD

TRANSFER DATA

OTI

Filter by name prefix only

Filter

Filter objects and folders

Show Live

☐	Name	Size	Type
☐	 zombie.jpg	71 KB	applicatio



- Include a screenshot of the output logs that show that the above image was blurred.

```
> * 2025-02-12 00:17:35.387 Analyzing zombie.jpg.  
> i 2025-02-12 00:17:35.787 vision.googleapis.com ...ImageAnnotator.BatchAnnotateImages images:annotate audit  
> * 2025-02-12 00:17:35.790 The image zombie.jpg was detected as inappropriate.  
> * 2025-02-12 00:17:35.853 Image zombie.jpg was downloaded to /tmp/tmpx6ixz2lj.  
> * 2025-02-12 00:17:41.750 Image zombie.jpg was blurred.  
> * 2025-02-12 00:17:41.915 Blurred image uploaded to: gs://jgafronsoredimages/zombie.jpg  
To view newer entries: Extend time by: 1 hour Edit time
```

- Why are there no items returned?

Because we subscribed to the pub AFTER the message was posted. We must have a active subscription to the pub BEFORE the message is posted. (I think)

- Take a screenshot of the output of the successful pull that includes the message and its **messageId**.

```
jgafron@pubsub:~$ gcloud pubsub subscriptions pull sub-$jgafron
```

DATA	MESSAGE_ID	ORDERING_KEY	ATTRIBUTES	DELIVERY_ATTEMPT	ACK_ID
Hello World...Again!	13688338025936540				UAYWLF1GSFE3GQhoUQ5PXiM_NSAoRRIBBk8CKF15ME0wQV18Dz4NGXJ9YXRjXkUIBkwFfF8PEQ1iXE5EB0mPkPPwV1dLWxAHCk1QfVdbGQhhX1pwA1E10Nn99MD1wjsbOX3R997WLS3216k8ZiI9XhJLLD5-NSNFQV5AEkw-FkRJUytDCypYEU4EISE-MD5FU0Q

- Take a screenshot showing the **messageIds** and messages sent

```
(env) jgafron@cloudshell:~/cs430-src/python-docs-samples/functions/v2/imagemagick (cloud-gafron-jgafron)$ python3 publisher.py
Enter a message to send: Hello world, a third time!!!!
Published 13688142139422125 to topic projects/cloud-gafron-jgafron/topics/my_topic
Enter a message to send: Hello world, a fourth time!!!
Published 13912192870118859 to topic projects/cloud-gafron-jgafron/topics/my_topic
Enter a message to send: Hello world....a fifth time!!!!
Published 13911917624933081 to topic projects/cloud-gafron-jgafron/topics/my_topic
Enter a message to send: Hello world, a sixth time!
Published 13688141614483946 to topic projects/cloud-gafron-jgafron/topics/my_topic
Enter a message to send: 
```

- Take a screenshot showing the same **messageIds** and messages received

```
jgafron@pubsub:~$ python3 subscriber.py
Received message 13912192870118859: 2025-02-12 08:52:15 (projects/cloud-gafron-jgafron/topics/my_topic) : Hello world, a fourth time!!!
Received message 13911917624933081: 2025-02-12 08:52:22 (projects/cloud-gafron-jgafron/topics/my_topic) : Hello world....a fifth time!!!!
Received message 13688141614483946: 2025-02-12 08:52:31 (projects/cloud-gafron-jgafron/topics/my_topic) : Hello world, a sixth time!

```